Amagi Framework — Master Version Control Document v1.1

Canonical Registry and Certification Baseline of the AMAGI Ecosystem

VERSION 1.1 — OCTOBER 2026

Alexey Mikhailovich Burlai AGIAM Technologies Ltd (In Formation) ORCID: 0009-0001-4679-5967
DOI: 10.5281/zenodo.20145223 (concept; a new version record DOI is issued upon publication)
License: CC BY-NC 4.0
Supersedes: Master Version Control Document v1.0 (2026-05-12) Status: Current — registry frozen as of 2026-10-26

1. Purpose

This document establishes the canonical versioning, dependency structure, lifecycle status, and certification baseline for all artifacts of the AMAGI ecosystem. Version 1.1 supersedes v1.0 and, in contrast to v1.0, registers the complete June 2026 document snapshot, introduces the status Baseline for Certification, transfers the superseded May registry and Master Parameter Sheet v2.0 to Archived/Obsolete status, and designates the IEEE TCAD manuscript as the Canonical Reference Implementation (Proof-of-Logic).

It resolves the chronological ambiguities of the May 2026 registry — which predated the June snapshot and therefore listed neither Framework v2.0, nor the v1.1 component specifications, nor the assurance documents — and defines the authoritative reference chain for regulatory, academic, and commercial contexts.

Scope. All published specifications, architectural definitions, licensing instruments, assurance documents, and supplementary materials of the AMAGI ecosystem.

Non-scope. Trade-secret implementation files (GDSII, SDC, calibration data) governed under separate NDA protocols per the Trade Secrets Notice. Implementation-internal working reports referenced for ⟨INPUT⟩ values are not registered here.

2. Configuration Baseline Concept

The ecosystem maintains a single configuration baseline composed of the **June 2026 snapshot** (§4) plus the correction-wave documents issued thereafter. Two structural rules govern interpretation:

2.1 Two-layer doctrine. Per MPS v2.1 §1.1, the canon of the class is the architectural layer (Layer A); implementation parameters belong to concrete instances (Layer I: RI-1, RI-2) and are recorded as dated evidence, not as canon. A bare number without a layer or instance tag is void for consistency purposes.

2.2 Precedence of the baseline. Where an archived or superseded document conflicts with the current baseline, the current baseline governs; archived documents are retained solely for historical traceability. Where a corpus document conflicts with the IEEE TCAD manuscript on parameters of RI-2, the manuscript governs (§6.2).

2.3 Terminology. Where any document conflicts with the Master Glossary v1.0 on terminology, the Master Glossary governs (precedence rules R1–R6 therein).

3. Document Registry

Status classes (extension of v1.0 §4): **Current / Baseline for Certification** (authoritative for new work; part of the certification baseline); **Current** (authoritative, not itself a certification-bound

item); **Superseded** (retained for historical traceability; reference only for legacy systems); **Archived/Obsolete** (retained solely for traceability; contains known defects listed in §5; shall not be cited as normative); **In preparation** (draft completed, pending author approval and Zenodo release); **Planned** (correction vehicle scheduled); **Related** (adjacent ecosystem; non-normative).

3.1 Core Architecture Layer

Document	Version	Date	Status	DOI / record
Amagi Framework Specification (AF)	v1.0	2025-11-22	Superseded	10.5281/zenodo.17679248
Amagi Framework (paper)	v1.3	2025-11-24	Superseded	10.5281/zenodo.17696607
Amagi Framework (AF-COMP)	v1.4	2025-12-07	Superseded — retained as RI-1 evidence (cited by MPS v2.1 §1.2)	10.5281/zenodo.17849595
Amagi Framework Specification (AF)	**v2.0**	2026-06	**Current / Baseline for Certification**; v2.1 corrigendum Planned (§5, item 3)	10.5281/zenodo.20753400
Amagi Open Specification (AOS)	v1.0	2025-12-09	Superseded (by Open Spec v1.1)	10.5281/zenodo.17861024
Amagi Open Specification	**v1.1**	2026-06	**Current / Baseline for Certification**; v1.2 corrigendum Planned (Glossary D-09)	10.5281/zenodo.20753403
IAA	v1.0	2025-12-15	Superseded	10.5281/zenodo.17938573
IAA	**v1.1**	2026-05-12	**Current / Baseline for Certification**	10.5281/zenodo.20145198
Trade Secrets Notice (TS-NOT)	v1.0	2025-12-10	Current	10.5281/zenodo.17886699

3.2 Component Specifications Layer

Document	Version	Date	Status	DOI / record
Amagi-MEM	v1.0	2025-11-22	Superseded	10.5281/zenodo.17679327
Amagi-MEM	**v1.1**	2026-06	**Current / Baseline for Certification**	10.5281/zenodo.20753402
Amagi-NIPU	v1.0	2025-11-22	Superseded	10.5281/zenodo.17682668
Amagi-NIPU	**v1.1**	2026-06	**Current / Baseline for Certification**	10.5281/zenodo.20753401

3.3 Assurance & Certification Layer

Document	Version	Date	Status	DOI / record
NIST AI RMF Profile	v1.0	2025-12-08	Superseded	10.5281/zenodo.17849725
NIST AI RMF Profile	**v1.1**	2026-06	**Current / Baseline for Certification**	10.5281/zenodo.20753404
HARA	v1.0	2026-06	Current — correction v1.1 In preparation (Wave 1 draft complete)	10.5281/zenodo.20753408
FMEDA	v1.0	2026-06	Current — correction v1.1 In preparation (Wave 1 draft complete); sole normative source of quantitative reliability parameters upon release of v1.1	10.5281/zenodo.20753410
Traceability Matrix (TM)	v1.0	2026-06	Current — correction v1.1 In preparation (Wave 1 draft complete)	10.5281/zenodo.20753412

Document	Version	Date	Status	DOI / record
Master Parameter Sheet (MPS)	v2.0	2026-05	**Archived/Obsolete** (effective upon publication of MPS v2.1; §5, item 1)	10.5281/zenodo.20145212
Master Parameter Sheet (MPS)	v2.1	2026-10	In preparation (Wave 1 draft complete)	new version of rec 20145212
Master Glossary	v1.0	2026-10	In preparation (Wave 2 draft complete)	new record
Master Version Control Document	v1.1	2026-10	**This document**	new version of rec 20145223

3.4 Legal & Governance Layer

Document	Version	Date	Status	DOI / record
AGIAM Founding Charter	v1.0	2025-11-26	Superseded	10.5281/zenodo.17719569
AGIAM Founding Charter	v1.1	2026-05-12	Superseded	10.5281/zenodo.20145094
AGIAM Founding Charter	**v1.2**	2026-06	**Current / Baseline for Certification**	10.5281/zenodo.20753872
S-APL	**v2.0**	2026-06	**Current / Baseline for Certification** — operative licensing instrument	10.5281/zenodo.20753406
License Notice (LIC-NOT)	v1.0	2025-11-22 *	Superseded — licensing function assumed by S-APL v2.0	10.5281/zenodo.17679248
Master Version Control Document	v1.0	2026-05-12	**Archived/Obsolete** (effective upon publication of this document; §5, item 2)	10.5281/zenodo.20145223

* License Notice (LIC-NOT): the document itself bears no printed date; the registry date is the publication date of the Zenodo record 17679248 that carries the file (2025-11-22).

3.5 Reference Implementation & Evidence Layer

Item	Designation	Status	Reference
IEEE TCAD manuscript	TCAD-2026-1011; submission ID f102b18b-4b04-42c1-ac80-d7ede7564009	Rejected 2026-09-22; new submission in preparation — **Canonical Reference Implementation (Proof-of-Logic)**, §6	—
RI-1 evidence set	VHDL prototype (2026-05): AF v1.4; MPS v2.0 §5/§18 (archived, read-only evidence); benchtop measurements	Retained as dated evidence for RI-1 only	MPS v2.1 §1.2
Verification package (Promela models, SPIN logs, property list, SVA)	—	**In preparation** — the corrected model scc_amagi_v3 and its verification logs are complete and frozen 2026-10-25 (closes TCAD R1 Major 3 and Open Problem 1); publication upon explicit owner approval (Wave 3; also closes TCAD R1 Major 10)	§9

3.6 Related Documents (Non-Normative)

Document	Version	Date	Status	DOI / record
Strategic Partnership Offer (Althea+AMAGI)	v1.0	2026-02-21	Related — excluded from the certification baseline; contains outdated DOI references (v1.0 §6); disposition pending author decision	10.5281/ zenodo.18718186
Althea corpus (SOP v2.0; CRISPR-TTP manuscript V3.0; Frontiers in Genetics 17:1727708)	—	2025–2026	Related — separate ecosystem; non-normative; the TTP term collision is resolved by Master Glossary v1.0 §8.3	10.5281/ zenodo.17676423 / 17780623 / 18605188

4. Certification Baseline (June 2026 Snapshot)

The following documents constitute the **Baseline for Certification** of the AMAGI Architectural Class:

#	Document	Version	Role in baseline
1	Amagi Framework Specification	v2.0 (2026-06)	Architectural canon; Protection Profile + integration specification
2	Amagi-NIPU	v1.1 (2026-06)	Processor-level enforcement; invariant realization; FSM chain-reset (§5.1)
3	Amagi-MEM	v1.1 (2026-06)	Memory-level enforcement; MAG/ATL/SHMF; Audit Diode (§3.3)
4	Amagi Open Specification	v1.1 (2026-06)	Unified component specification; ADR register
5	S-APL	v2.0 (2026-06)	Operative licensing instrument
6	NIST AI RMF Profile	v1.1 (2026-06)	Regulatory mapping
7	AGIAM Founding Charter	v1.2 (2026-06)	Governance
8	IAA	v1.1 (2026-05-12)	Architectural class membership
9	Trade Secrets Notice	v1.0 (2025-12-10)	Confidentiality boundary
10	HARA / FMEDA / TM	v1.0 (2026-06) → v1.1	Assurance documents; v1.1 corrections (Wave 1) inherit baseline status upon publication
11	MPS v2.1; Master Glossary v1.0; VC v1.1	2026-10	Freeze instruments; inherit baseline status upon publication

Certification is performed per implementation instance by accredited bodies; architectural alignment of the baseline does not itself constitute a certified product (MPS v2.1 §15).

5. Superseded Documents

The following documents are transferred to **Archived/Obsolete** status. Published Zenodo records are immutable and cannot be deleted; the status is semantic, enforced through this registry, versioned DOIs, and the precedence rules of §2.

5.1 Master Parameter Sheet v2.0 (rec 20145212). Status: Archived/Obsolete, effective upon publication of MPS v2.1. Grounds: (a) factual defects — crypto primitive «SPHINCS+-256s / 29 792 B» instead of SLH-DSA-SHA2-128f (~7 856 B); classification of FIPS 205 SLH-DSA as stateful; device part Artix-7 100T instead of XC7A35T; power 1.5 W instead of < 1.3 W (RI-2 parameters); (b) superseded reliability figures — PFDavg 3.6×10^{-4} / 1.8×10^{-4}, SFF 92%, subsystem Type A (contradict FMEDA v1.0/v1.1); (c) terminology — Memory Architecture Grid, Access Time Latency, Temporal Triad Parameters, «Stateful» classification (Master Glossary v1.0 D-04, D-05, D-08, D-10, D-14). The v2.0 record is retained solely as RI-1 evidence (§3.5) and for historical traceability; any parameter conflict with MPS v2.1 is resolved in favour of MPS v2.1.

5.2 Master Version Control Document v1.0 (rec 20145223). Status: Archived/Obsolete, effective upon publication of this document. Grounds: registry frozen at 2026-05-12; does not contain the June 2026 snapshot; lists MPS as «v1.0» while the record carries v2.0; LIC-NOT DOI marked [Pending] although published under rec 17679248. A proposal to resolve these defects by rolling the ecosystem baseline back to Framework v1.4 / Open Spec v1.0 was **considered and rejected**; the accepted remedy is this v1.1 registry, which legitimizes the June snapshot as the new canon. Dependency graph and known-issues table of v1.0 are hereby superseded by §3 and §10.

5.3 General supersession clause. All prior versions of the documents registered in §3 not listed as Current are Superseded. Superseded documents remain referenceable for legacy-system traceability only; they shall not be cited as normative in new work.

6. Canonical Reference Implementation (Proof-of-Logic)

6.1 Designation. The manuscript «AMAGI: A Formal Verification Framework for Hardware-Enforced AI Safety», submitted to IEEE TCAD (submission ID **f102b18b-4b04-42c1-ac80-d7ede7564009**; internal designation **TCAD-2026-1011**; **rejected 2026-09-22, new submission in preparation**; referenced as a normative source by NIST Profile v1.1 and FMEDA v1.0), is designated the **Canonical Reference Implementation (Proof-of-Logic)** of the AMAGI Architectural Class, instantiated as **RI-2** (SystemVerilog, 467 lines, current reference implementation per MPS v2.1 §1.2).

6.2 Discrepancy rule. Any parameter conflict between the manuscript and older corpus drafts concerning RI-2 is resolved **in favour of the manuscript**. Named applications of this rule: FPGA part **XC7A35T** (not Artix-7 100T); power envelope **< 1.3 W** (not 1.5 W); crypto primitive **SLH-DSA-SHA2-128f** (not SPHINCS+-256s). Family-level references to «Xilinx Artix-7» in AF v2.0/ FMEDA v1.0/NIST Profile v1.1 remain valid as class-level statements.

6.3 Immutable evidence. The following results of the manuscript are fixed and shall not be restated with different values in any corpus document: SPIN state space **25 927 680** states; response latencies **12–35 µs** (TCB-only envelope); SFF **93%** (system-level); prototype conditions: **post-synthesis evidence (Xilinx Vivado 2023.2), single XC7A35T device** (proof-of-logic scope); a board-level measurement campaign has not been performed and is designated for Phase 2. RI-1 figures (VHDL ~340 lines; 2.43×10^6 states; 100 MHz) remain valid exclusively as RI-1 attributes.

6.4 Bibliographic maintenance. The manuscript was rejected on 2026-09-22; a new submission with a response letter summarizing the fundamental changes is in preparation. Upon acceptance of the new submission, its bibliographic data will be fixed by erratum to this document. The designation as Canonical Reference Implementation (Proof-of-Logic) is independent of the review outcome.

7. Binding Configuration Decisions

The following decisions of the 2026 consistency freeze are binding for all corpus documents and supersede any conflicting statement in archived documents.

ID	Decision	Normative location
D1	Two-layer model: Layer A (architecture, canon) / Layer I (instances); RI-2 = current reference implementation; bare numbers read as RI-1 attributes unless tagged	MPS v2.1 §1.1–1.2
D2	Integrity Capability Formula: **SIL 2 @ 1oo1 (HFT=0) — reference configuration, current; SIL 3 @ 1oo2 (HFT=1, β=0.10) — enhanced configuration, Phase 2 target (2027)**; binding constraint is the architectural constraint per IEC 61508-2:2010 Table 2 (Type B, SFF)	MPS v2.1 §3.1; HARA v1.1; FMEDA v1.1 §7–8

ID	Decision	Normative location
D3’	**T1 (proof test interval) = 8 760 h ≠ T_M (mission/design life) = 87 600 h**; the «1 year vs 10 years» contradiction is terminological, not substantive	MPS v2.1 §3.3; FMEDA v1.1 §7.1; AF v2.1 corrigendum
D4	INV-004 policy update path = Dual-Tier bank switching per **ADR-002** (active bank + staging bank; atomic activation; no rollback bypassing SCC)	MPS v2.1 §6; Open Spec v1.1; TM v1.1 SR-15
D5	Isolation profiles for INV-001: **S** (subsystem, reference, current) / **D** (device-level, Phase 2 2027)	MPS v2.1 §1.3
CCR-01	Terminology unification per Master Glossary v1.0: **ATL = Access Trigger Layer, MAG = Memory Access Graph** (and MPE, CRTM, TTP canon); rename-only fix in HARA v1.1 and FMEDA v1.1 — **no change of architectural roles**; the audit function (Audit Diode) is embedded in the TTP Engine/ATL per MEM v1.1 §3.3 and is not delegated to SHMF	Master Glossary v1.0 §§7–8
CCR-02	Status of liveness property L04 (latch-up): **«Hardware mitigation implemented (FSM chain-reset, NIPU v1.1 §5.1, safety invariants preserved); formal re-verification of the corrected Promela model completed 2026-10 (scc_amagi_v3: L01–L04 verified, FA1/FA2 verified — Wave 3 package, publication pending); Isabelle/HOL formalization remains on the roadmap»**. The former «Phase 2» qualifier is discharged for the Promela model by the completed re-verification and refers now exclusively to the Isabelle/HOL formalization	TM v1.1; NIPU v1.1 §5.1; Wave 3 results_v3.md (T-002, 2026-10-25)

8. Versioning and Publication Rules

8.1 Semantic versioning (inherited from v1.0 §5): MAJOR (X.0) — invariant-breaking changes, new architectural class definitions; MINOR (X.Y) — additive features, new components, expanded compliance coverage; PATCH (X.Y.Z) — errata, clarifications, non-normative corrections. Document-specific rules of v1.0 §5 (AF-SPEC major version aligns with IAA invariant set; MEM/NIPU reference current AOS; IAA version locked to invariant set; AGIAM-CH version locked to governance structure) remain in force.

8.2 Zenodo mechanics: each published record is immutable; a new version is released as a new record with a new DOI (the concept DOI continues to resolve to the latest version); deletion of published records is not possible — defect handling is exclusively by new versions and registry status.

8.3 Release protocol: draft → diff review by the author → explicit approval per record → sandbox rehearsal → production release → registry update by a new version of this document. No document enters the baseline without an explicit approval record.

8.4 Freeze instruments. The Master Glossary governs terminology (precedence R1–R6); this document governs statuses and baselines; the FMEDA (v1.1 upon release) is the single normative source of quantitative reliability parameters; the MPS (v2.1 upon release) is the single consolidated parameter reference with layer/instance tagging.

9. Correction Wave Plan (Provisional)

Wave	Contents	Status
Wave 1	MPS v2.1; HARA v1.1; FMEDA v1.1; TM v1.1	Drafts complete (2026-10), pending author approval
Wave 2	Master Glossary v1.0; VC v1.1 (this document)	Drafts complete (2026-10), pending author approval

Wave	Contents	Status
Wave 3	AF v2.1 corrigendum («10-year design life» wording, D3'; firmware-signing wording per §10, item 8); Open Spec v1.2 corrigendum (TTP acronym row, Glossary D-09); verification package release (Promela/SPIN/SVA — closes TCAD R1 Major 10); FMEDA metadata correction (keywords: IEC 61508-2, not ISO 26262)	Planned
Non-Zenodo	Public site conformance: SIL formula (D2), SPIN badge with CEX caveat, INV-004 card (ADR-002), removal of «peer-reviewed» and absolute-protection claims	Planned

The wave plan is provisional and subject to author adjustment; the registry (§3) governs over the plan in case of divergence.

10. Known Issues & Resolutions

#	Issue	Affected documents	Status	Resolution target
1	Registry missing June 2026 snapshot; MPS version label mismatch	VC v1.0	**Resolved**	This document
2	LIC-NOT DOI [Pending]	VC v1.0	**Resolved**	DOI 10.5281/zenodo.17679248; status Superseded (§3.4)
3	«10-year TI» wording	AF v2.0	Open	AF v2.1 corrigendum (D3')
4	«Stateful hash-based signatures for firmware» wording vs firmware verification by ML-DSA-44 (Open Spec v1.1 §8); SP 800-208 cited in bibliography	AF v2.0	Open — author to confirm scope (stateful XMSS/LMS migration context vs residual error)	AF v2.1 corrigendum
5	TTP expansion in acronym table	Open Spec v1.1	Open (Glossary D-09; R1 governs meanwhile)	Open Spec v1.2
6	Outdated DOI references in external documents	Strategic Partnership Offer v1.0	Open (v1.0 §6 carried forward)	Pending author disposition (§3.6)
7	Parameter and terminology defects of MPS v2.0	MPS v2.0	**Resolved by supersession**	MPS v2.1 (Wave 1); §5.1
8	Term drift MAG/ATL/TTP/MPE/CRTM/SLH-DSA across corpus	HARA v1.0, FMEDA v1.0, MPS v2.0, Open Spec v1.1	**Resolved** by Master Glossary v1.0 + CCR-01; per-document fixes tracked in Glossary §7	Wave 1/3 vehicles
9	SIL 2 vs SIL 3; TI 1 vs 10 years; L04 status divergence	AF v2.0, HARA v1.0, FMEDA v1.0, TM v1.0	**Resolved** (D2, D3', CCR-02)	Wave 1 documents
10	RI-2 verification package not public	Ecosystem	Open	Wave 3 (R1 Major 10)
11	§3 registry dates of five v1.0 records (MEM, NIPU, NIST, AOS, LIC-NOT) did not match the Zenodo record dates	VC v1.0	**Resolved**	Dates corrected in this document (§3.1–3.4); dating rule: registry dates are Zenodo record publication dates

#	Issue	Affected documents	Status	Resolution target
12	Manuscript status printed as «Under review»; «n = 75 @ 25 °C» stated in §6.3 without a primary source	VC v1.1 draft	**Resolved**	Status corrected to «Rejected 2026-09-22; new submission in preparation» (§3.5, §6.1, §6.4, [16]); «n = 75» withdrawn — prototype scope stated per post-synthesis evidence (§6.3; consistent with FMEDA v1.1 and MPS v2.1)
13	INV-005 coverage count: §7 of HARA v1.0 lists six hazard IDs against a stated count of 5; TM v1.0 §5 records 5	HARA v1.0, TM v1.0	**Resolved conservatively**	HARA v1.1 aligns the §7 list to the stated count 5 (consistent with Exec Summary, §10 and TM); recount of Table 5 deferred to the next HARA revision

11. Change Log

Version	Date	Author	Changes
0.1	2026-05-12	A.M. Burlai	Initial draft, registry compilation
1.0	2026-05-12	A.M. Burlai	Reviewed, all DOIs verified, dependency graph completed
1.1	**2026-10-26**	A.M. Burlai (consistency freeze)	Full June 2026 snapshot registered; Baseline for Certification established (§4); MPS v2.0 and VC v1.0 → Archived/Obsolete (§5); TCAD manuscript designated Canonical Reference Implementation with paper-precedence rule (§6); binding decisions D1–D5, CCR-01, CCR-02 recorded (§7); release protocol and wave plan added (§8–9); known issues extended (§10)

12. References

1. Burlai, A.M. (2026). Amagi Framework Specification v2.0. DOI: 10.5281/zenodo.20753400
2. Burlai, A.M. (2026). Amagi Open Specification v1.1. DOI: 10.5281/zenodo.20753403
3. Burlai, A.M. (2026). Amagi-NIPU v1.1. DOI: 10.5281/zenodo.20753401
4. Burlai, A.M. (2026). Amagi-MEM v1.1. DOI: 10.5281/zenodo.20753402
5. Burlai, A.M. (2026). AMAGI NIST AI RMF Profile v1.1. DOI: 10.5281/zenodo.20753404
6. Burlai, A.M. (2026). Amagi HARA v1.0. DOI: 10.5281/zenodo.20753408
7. Burlai, A.M. (2026). Amagi FMEDA v1.0. DOI: 10.5281/zenodo.20753410
8. Burlai, A.M. (2026). Amagi Traceability Matrix v1.0. DOI: 10.5281/zenodo.20753412
9. Burlai, A.M. (2026). Amagi-SAPL v2.0. DOI: 10.5281/zenodo.20753406
10. Burlai, A.M. (2026). AGIAM Founding Charter v1.2. DOI: 10.5281/zenodo.20753872
11. Burlai, A.M. (2026). IAA v1.1. DOI: 10.5281/zenodo.20145198
12. Burlai, A.M. (2025). AMAGI Framework Trade Secrets Notice v1.0. DOI: 10.5281/zenodo.17886699
13. Burlai, A.M. (2026). Master Parameter Sheet v2.0 (Archived/Obsolete). DOI: 10.5281/zenodo.20145212
14. Burlai, A.M. (2026). Master Version Control Document v1.0 (Archived/Obsolete). DOI: 10.5281/zenodo.20145223
15. Burlai, A.M. (2026). AMAGI Master Glossary v1.0. [new record — DOI upon publication]
16. Burlai, A.M. (2026). AMAGI: A Formal Verification Framework for Hardware-Enforced AI Safety. IEEE TCAD submission f102b18b-4b04-42c1-ac80-d7ede7564009 (TCAD-2026-1011), rejected 2026-09-22; new submission in preparation.